

Engineering Technology - Short Note Part 07

Electricity Generation, Transmission & Distribution

Original Content by: Iresh Sadeepana (Tech Hub) | English Translation & Technical Summary

1. Introduction to Power Generation

Electricity generation is broadly classified into two categories based on the energy sources used:

- **Renewable Energy Sources** Solar, Hydro, Wind, Geothermal, Biomass, Ocean Waves, Tidal Energy
- **Non-Renewable Energy Sources** Petroleum Oils (Petrol, Diesel, Kerosene), Coal, Natural Gas, Nuclear Energy

2. Renewable Power Generation

A. Solar Energy

Solar energy is harvested using Photovoltaic (PV) systems. Systems are divided into two main categories:

1. **Stand-Alone Solar PV Systems:** Solar Panel → Charge Controller → Battery Bank → Inverter → AC Load (Home).
2. **Grid-Connected Solar PV Systems:** Solar Panel → Grid-Tied Inverter → AC Breaker → Utility Meter / Grid.

Solar Thermal Power Generation: Uses solar absorbers, heat exchange/storage units, and secondary fluid loops (oil/water) to generate high-pressure steam driving a steam turbine.

B. Wind Energy

Wind energy uses wind turbines (windmills) to convert kinetic energy into electrical energy:

- **Main Components of a Wind Turbine:** Rotor Blades, Hub, Pitch Mechanism, Low-Speed Shaft, Gearbox, Brake System, High-Speed Shaft, Generator, Controller, Anemometer (Wind Speed Measurement), Wind Vane (Wind Direction Measurement), Nacelle, Yaw Motor, Yaw Drive, Tower.
- **Operation:** Wind rotates large turbine blades → Gearbox increases rotational speed → Generator produces electricity → Transformer steps up voltage for grid transmission.

C. Biomass & Biogas Energy

Biomass chemical energy is utilized in two ways:

1. **Biomass Combustion:** Direct burning of biomass → Boiler heats water to steam → Steam turbine rotates electric generator.
2. **Biogas Generation (Anaerobic Digestion):** Organic/animal waste decomposes under low-oxygen conditions → Produced biogas fuels Combined Heat and Power (CHP) units to generate electricity and district heating.

D. Hydroelectric Energy

Power generated in a hydroelectric power plant depends on the water mass (m), gravitational acceleration (g), and water head (h).

Potential Energy: $E = mgh$

Water Head: The vertical elevation difference between the reservoir surface level and the turbine discharge point. Higher head produces greater potential energy.

Classification by Water Head Height (*H* in meters):

- Low Head Plants: *H* ≤ 15 m
- Medium Head Plants: 15 m < *H* ≤ 70 m
- High Head Plants: 70 m < *H* ≤ 250 m
- Very High Head Plants: *H* > 250 m

Classification by Capacity:

- Micro/Very Low Capacity: Up to 0.1 MW
- Small Capacity: 1 MW - 10 MW
- Medium Capacity: Up to 100 MW
- High Capacity: Above 100 MW

Key Components of a Hydro Power Plant:

- **Catchment Area:** Region receiving rainfall that supplies water to the reservoir.
- **Reservoir & Dam:** Impounds water collected from catchment area.
- **Pressure Tunnel:** Conveys water to the power station; tapering diameter increases fluid pressure.
- **Surge Tank:** Absorbs sudden pressure spikes (water hammer effect) when generator valves close abruptly, preventing pipe rupture.
- **Penstock:** High-gradient steel pipes that narrow down to build up water pressure.
- **Turbines:** Classified into *Impulse Type* (Pelton Wheel, Cross-Flow) and *Reaction Type* (Francis, Propeller/ Kaplan).

Turbine Type	Mechanism	Application / Head Condition
Francis Turbine (Reaction)	Utilizes pressure and kinetic energy	Medium Head Plants
Kaplan / Propeller (Reaction)	Adjustable/Fixed blades	Low Head, High Flow Rate Plants
Pelton Wheel (Impulse)	High-velocity water jets via nozzles	High Head Plants

E. Ocean & Geothermal Energy

- **Ocean Wave & Tidal Energy:** Harnesses potential energy difference between high/low tides or kinetic energy of wind-driven ocean waves.
- **Geothermal Power Generation:** High-pressure cold water is injected into hot underground rock formations via production/injection wells → Hot water/steam returns to surface → Heat exchanger transfers energy to secondary fluid → Vapor drives steam turbine to generate power.

3. Non-Renewable Power Generation

Non-renewable energy utilizes fossil fuels (Petroleum, Coal, Natural Gas) and Nuclear Energy:

- **Petroleum Thermal Plants:** Oil boiler generates steam to drive steam turbines, or combustion gases directly power gas turbines and internal combustion engines.
- **Coal Power Plants:** Coal combustion → Boiler heats water to high-pressure steam → Steam turbine → Condenser cools exhaust steam back to water loop.
- **Natural Gas Power Plants:** Air compressor + Combustion chamber → Gas turbine drives generator + Exhaust heat produces secondary steam driving a secondary steam turbine (Combined Cycle).
- **Nuclear Power Plants:** Nuclear reactor vessel (with Control Rods & Pressurizer) undergoes fission → Heat generates steam in steam generator → Steam drives turbine.

4. Electricity Transmission & Distribution

Structure of the Electrical Grid:

- **Generation Voltage:** Typically **11 kV**, **12.5 kV**, or **13.5 kV**.
- **Step-Up Transmission (Generation Substation):** Voltage stepped up to **132 kV** or **220 kV**.
- **Grid Substation:** Steps down from **220 kV / 132 kV** to **33 kV** (Supplies large industrial bulk customers).
- **Primary Substation:** Steps down from **33 kV** to **11 kV** (Supplies large commercial customers).
- **Distribution Substation:** Steps down from **11 kV / 33 kV** to **400 V** (3-phase, 4-wire) / **230 V** (1-phase) for residential and normal customers.

Advantages of High-Voltage Power Transmission:

1. **Reduced Potential Drop:** Stepping up voltage reduces current proportionally for given power, minimizing line voltage drop.
2. **Lower Power Losses:** Power loss ($P = I^2R$) decreases exponentially as transmission current (I) is reduced.
3. **Economical Conductor Size:** Lower current requires smaller cross-sectional wire area, drastically reducing conductor material costs.
4. **3-Wire Delta Transmission:** Step-up transformer secondary windings are Delta-connected, allowing power transmission using 3 conductors instead of 4, saving material costs.

Distribution Substation Equipment:

- HT (High Voltage) Terminals & LV (Low Voltage) Terminals
- Insulator Disks
- Drop Down Lift Off (DDLO) Fuses
- Surge Arrestors (Protection against lightning surges)
- Panel Board with High Rupturing Capacity (HRC) Fuses & Metering Instruments

5. Electrical Safety & Protection

Types of Electric Shock Contact:

- **Direct Contact:** Direct touch with an exposed energized conductor/phase wire.
- **Indirect Contact:** Contact with metallic conductive parts that have become live due to an internal fault or insulation failure.

Factors Affecting Severity of Electric Shock:

- 1. Path of current flow through the body and body resistance.
- 2. Magnitude and nature (AC/DC) of current.
- 3. Duration of exposure/contact with the live circuit.

Current Level (Amperes)	Human Body Physiological Response
0.001 A (1 mA)	Tingling sensation felt on skin.
0.009 A (9 mA)	Muscular contraction; nervous system affected by supply frequency.
0.030 A (30 mA)	Breathing difficulty; severe danger of non-recovery after 1 minute.
0.1 - 0.2 A (100 - 200 mA)	Ventricular fibrillation of heart capillaries; instant death risk.

Note on Exposure Time: Currents below 10 μ A flowing through the body for longer durations are generally harmless, whereas higher currents cause severe damage even in fractions of a second.

Emergency First-Aid Protocol for Electric Shock Victims:

Step 1: Immediately disconnect power supply / turn off main breaker.
Step 2: Safely move victim away from the hazard source (Extinguish fire if present).
Step 3: Call emergency medical assistance / doctor immediately.
Step 4: Check victim's state & injuries (Breathing, pulse, shock condition).
Step 5: Perform CPR/Artificial respiration if breathing stopped; elevate legs if in shock; position sideways if unconscious.
Step 6: Transport victim to hospital for professional medical evaluation.